

Práctica 5.1

15-Sept-2026

Diseño de un controlador para un sistema de segundo orden

Ecuaciones Principales

$$V_e(t) = Ri_1 + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = RI_1(s) + Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = RI_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento Algebraico

$$V_e(s) = (Ls + 2R)I_1(s) - (Ls + R)I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$V_s(s) = (R + \frac{1}{Cs}) I_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR^2s^2 + (CR^2 + L)s + R}{3(LR^2s^2 + (5CR^2 + L)s + 2R)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(R + \frac{1}{Cs}) I_2(s)}{(Ls + 2R) \left[\frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s) \right] - (Ls + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(\frac{R(s+1)}{Cs}) I_2(s)}{(Ls + 2R) \left[\frac{1}{Ls + R} \left(\frac{Ls^2 + 3RCs + 1}{Cs} \right) I_2(s) \right] - (Ls + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(\frac{R(s+1)}{Cs}) I_2(s)}{(Ls + 2R) \left[\frac{Ls^2 + 3RCs + 1}{CLs^2 + RCS} \right] I_2(s) - (Ls + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(\frac{R(s+1)}{Cs}) I_2(s)}{\left[\frac{L^2Cs^3 + 3LRCs^2 + Ls}{CLs^2 + RCS} + \frac{2L(Rs^2 + 6R^2Cs + 2R)}{CLs^2 + RCS} - (Ls + R) \right] I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(\frac{R(s+1)}{Cs}) I_2(s)}{\left[\frac{L^2Cs^3 + 3LRCs^2 + Ls + 2L(Rs^2 + 6R^2Cs + 2R) - (L^2s^3 + RLCs^2 + CLs + R^2Cs)}{CLs^2 + RCS} \right] I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(\frac{R(s+1)}{Cs}) I_2(s)}{\left[\frac{3RLCs^2 + (5R^2C + L)s + 2R}{CLs^2 + RCS} \right] I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3(LRs^2 + (5R^2C + L)s + 2R)}$$

* Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{V_s(s)}{V_e(s)} \right) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{RCLs^2 + (R^2C + L)s + R}{3RLCs^2 + (5R^2C + L)s + 2R} \right)$$

$$e(s) = 1 - \frac{R}{2R} = 0.5$$

$$e(s) = 0.5$$

$$e(t) = 0.5V$$

* Estabilidad en lazo abierto

$$3RLCs^2 + (5R^2C + L)s + 2R = 0$$

$$10.048 \times 10^{-3} s^2 + 371.25 \times 10^3 s + 30 \times 10^3 = 0$$

$$R = 15k\Omega$$

$$L = 680\mu H$$

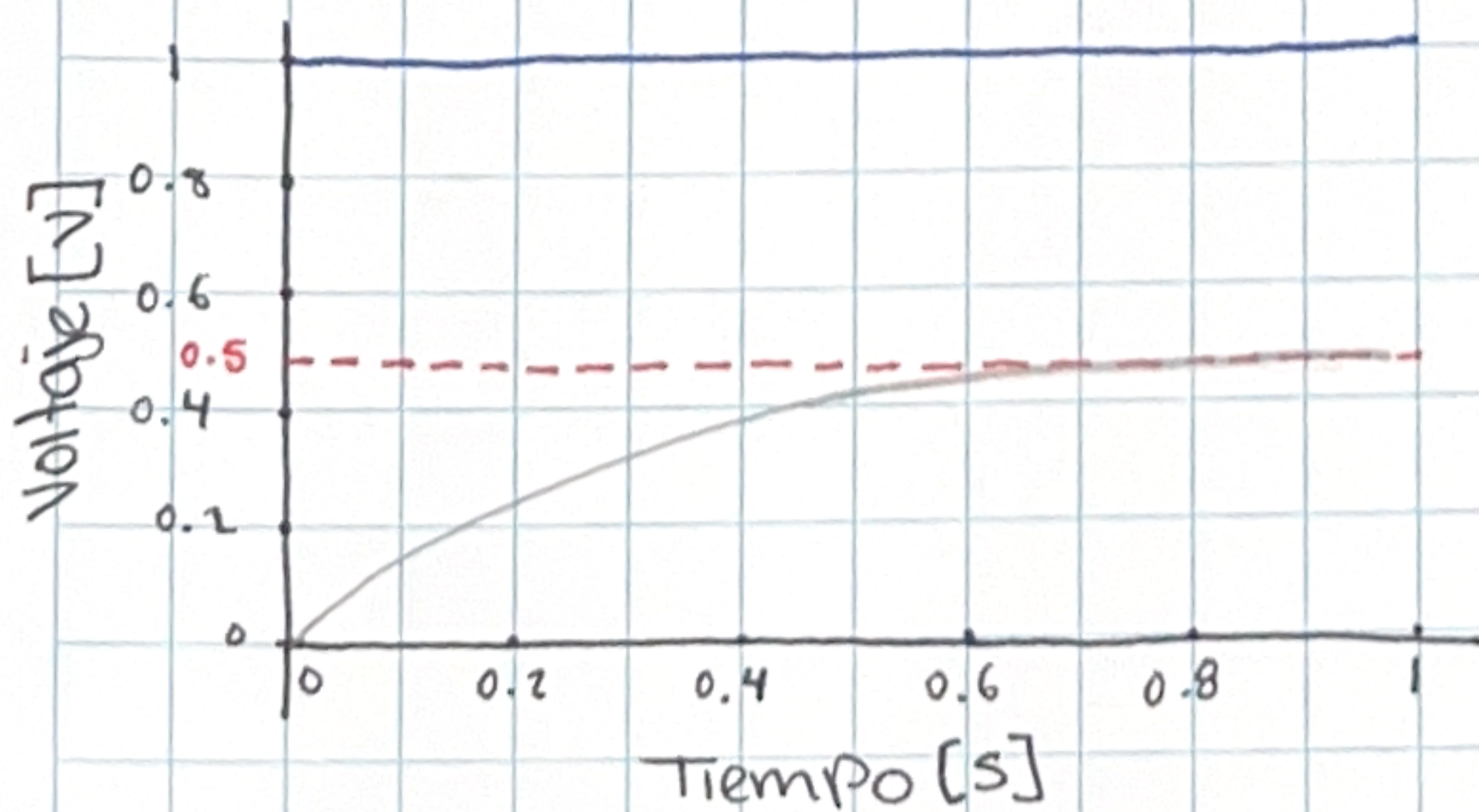
$$C = 330\mu F$$

$$\lambda_1 = -80.80 \times 10^{-3}$$

$$\lambda_2 = -36.76 \times 10^6$$

El sistema es estable y tiene una respuesta sobreamortiguada.

Bosquejo



- $V_e(t)$

--- Error en estado estacionario

Modelo de ecuaciones integro-diferenciales
 El modelo se expresa despejando las variables dependientes, es decir, las corrientes.

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$i_2(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_2(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$